



December 2025

(For ME)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

14. In normalizing of hypoeutectoid steel, the component is heated to a temperature above
 A) 535°C B) 610°C
 C) 800°C D) 910°C
15. Which of the following processes will one use on hardened steel to reduce brittleness?
 A) Annealing B) Normalizing
 C) Spheroidizing D) Tempering
16. For steel, which one of the following properties can be enhanced upon annealing?
 A) Hardness B) Toughness
 C) Ductility D) Resilience
17. In which of the following stages do the deformation rate increases and causes failure?
 A) Transient creep stage B) Constant creep stage
 C) Fracture stage D) Steady stage creep stage
18. Which of the following is costliest among the non ferrous materials?
 A) Magnesium B) Aluminum
 C) Titanium D) Copper
19. Which of the following is the lightest among the following?
 A) Magnesium B) Aluminum
 C) Titanium D) Copper
20. Composite can be classified on the basis of:
 A) Matrix type B) Reinforcement constituent
 C) Matrix type & Reinforcement constituent D) None of the mentioned

PART - B: DESCRIPTIVE ANSWER QUESTIONS.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Draw the unit cell of a body-centered cubic crystal and face-centered cubic crystal.	04	L*1	01	01
b)	Define coordination number in crystal structure, mention coordination number for SC, BCC and FCC.	04	L1	01	01
c)	What is a solid solution? Explain its types with suitable examples.	08	L1	02	01
2. a)	What are surface defects? Explain the different types of surface defects with neat diagrams.	08	L2	01	01
b)	What is cast metal structure? With neat diagram explain cast structure in metals.	04	L1	02	01
c)	Explain the concept of solidification of metals with a neat sketch.	04	L1	02	01
3. a)	Explain the Bravais lattices belonging to the Tetragonal and Orthorhombic crystal systems. Draw neat diagrams to illustrate their unit cell structures.	08	L2	01	01
b)	Two metals A and B have their melting points at 900°C and 800°C respectively. The alloy pair forms a eutectic at 600°C of composition 60% B and 40% A. A and B have unlimited mutual liquid solubilities. Their solid solubilities are as follows. 10% B in A at 600°C and 5% B in A at 0°C 8% A in B at 600°C and 4% A in B at 0°C Assume the liquidus, solidus and solvus lines to be straight. No solid state reactions or any intermediate phase changes occur in the series, i) Draw the phase diagram for the series and label all salient temperatures, compositions and regions. ii) Find the room temperature (25°C) structure of any alloy of composition 60% A and 40% B, with respect to the number, type, extent and composition of the phases.	08	L3	02	01
Unit – II					
4. a)	What is T.T.T diagram? Explain with a neat diagram the Martensitic transformation of Austenite.	08	L2	03	01

b)	Explain Annealing and Martempering heat treatment processes with neat diagram.	08	L2	04	01
5. a)	Explain the Jominy End-Quench Test with a neat sketch. How is this test useful in determining the hardness and hardenability of steels.	08	L2	03	01
b)	Write a note on Surface treatments. Explain the Induction hardening process with neat sketch.	08	L2	04	01
6. a)	Draw a neat sketch of Fe-Fe ₃ C equilibrium diagram and briefly explain.	08	L2	03	01
b)	Draw with a separate diagram and explain Tempering and Normalizing heat treatment processes.	08	L2	04	01

Unit – III

7. a)	Explain the importance of the matrix and reinforcement in composite materials.	04	L1	05	01
b)	List linear and non- linear properties of materials.	04	L1	05	01
c)	Explain the different types of fatigue cycles with a neat diagram.	04	L1	05	01
d)	Explain effect of alloying elements in Steels.	04	L1	05	01
8. a)	Explain with the sketch R.R. Moore's fatigue test and draw S-N curve.	08	L2	05	01
b)	Explain the role of matrix and reinforcement in composite materials.	04	L1	05	01
c)	Differentiate Malleable cast iron Grey cast iron with neat diagram.	04	L1	05	01

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
